

ISYS90045 Professional IS Consulting

Seminar 9 Professional Analysis Hypothesis Driven Consulting

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Learning Objectives

Professional Analysis

- Explore how Consultants solve problems?
- Introduce the concept of hypothesis based problem solving
- Define and understand a hypothesis
- Define and understand hypothesis driven consulting

How does a Consultant solve problems?

Acknowledgement: Christina Parkes

Hypothesis Driven Consulting

A hypothesis is an idea or theory, often based on limited data, which is typically the beginning of a thread of further investigation to prove, disprove or improve the hypothesis through facts and empirical data.



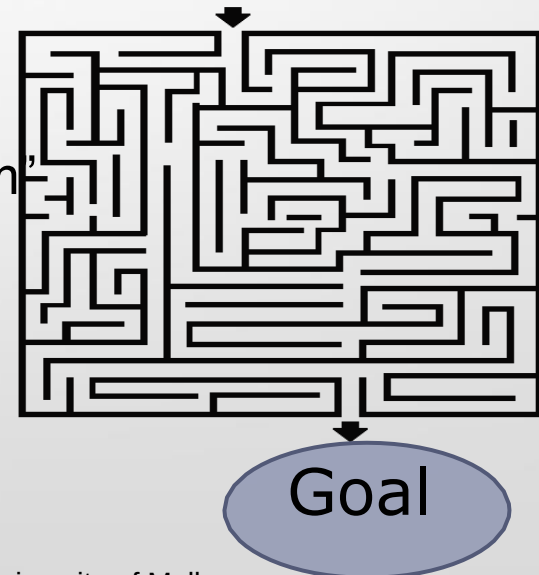
What is a Hypothesis?

A hypothesis-driven consulting approach is:

It's more than educated guessing; this is how consultants smartly break down complex or ambiguous problems, and quickly start driving towards an answer.

At its heart is the scientific method – used for hundreds of years by scientists and thinkers to prove their ideas using evidence. Going as far back as Aristotle, Bacon, Descartes...

- ▶ Also known as “The labyrinth problem”

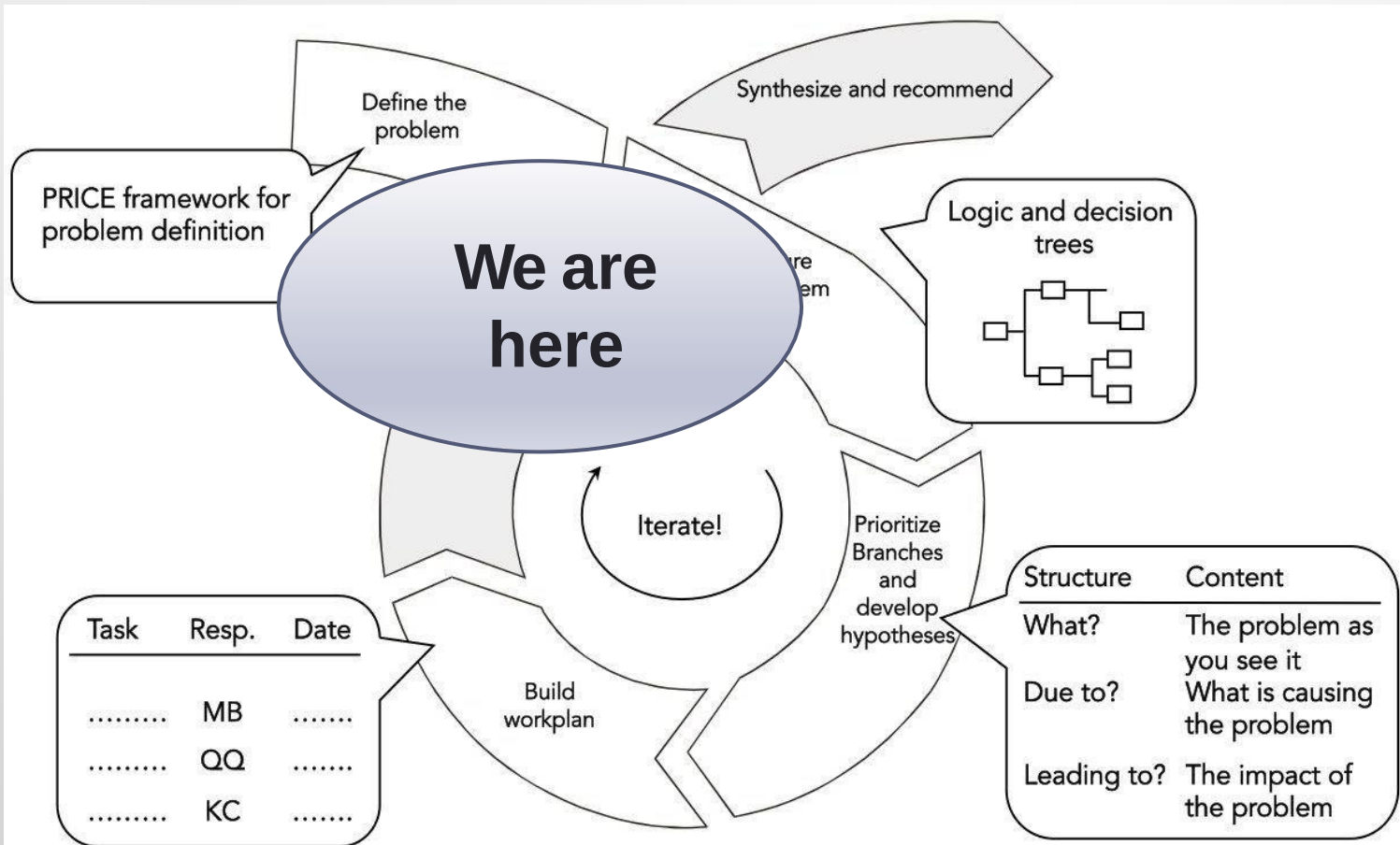


How do we use a hypothesis?

What does it mean to be hypothesis-driven?

- ▶ A hypothesis-driven consultant develops a hypothesis of the answer to a problem early
- ▶ Focuses on **testing** and **revising** that hypothesis over the course of the engagement
- ▶ This is also known as **having a perspective or Point of View (POV)**
- ▶ The hypothesis is used to **inform** what **data** needs to be collected, **analyses** will be performed, and **insights** must be gained in order to arrive at a final answer
- ▶ In contrast;
 - ▶ Operational problems solving has a linear process to it and focusses of root causes with an outcome determined before it's implemented

Process for Hypothesis Driven



Examples

Some common forms of “hypothesis-driven consulting questions” are:

- ? Why do you think that X happens?
- ? What are some hypotheses you can think of that explain ABC?
- ? What are some possible reasons for XYZ?

For Example:

Why do you think our staff turnover is 10%? When it used to be 2%

What reasons explain why our staff unproductively browse the web?

How do we use a hypothesis?

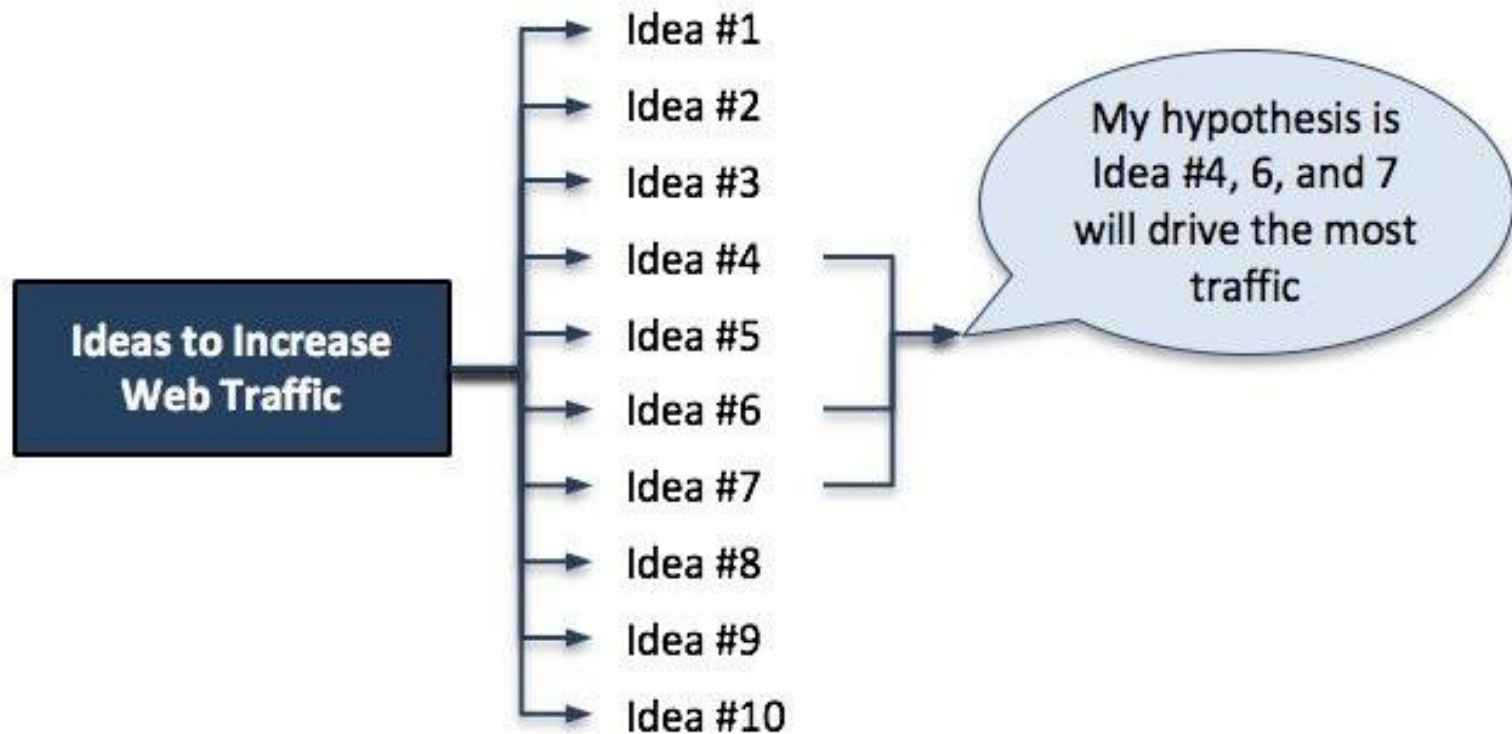
1. The first step in being hypothesis-driven is to focus on the highest potential ideas and theories of how to solve a problem or realize an opportunity.

An example of being hypothesis-driven:

You own a website, and you brainstorm ten ideas to improve web traffic, but you don't have the budget to execute all ten ideas. The first step in being hypothesis-driven is to prioritize the ten ideas based on how much impact you hypothesize they will create.

Example: Hypothesis process

Hypothesis Driven Prioritization



Hypothesis Driven Consulting

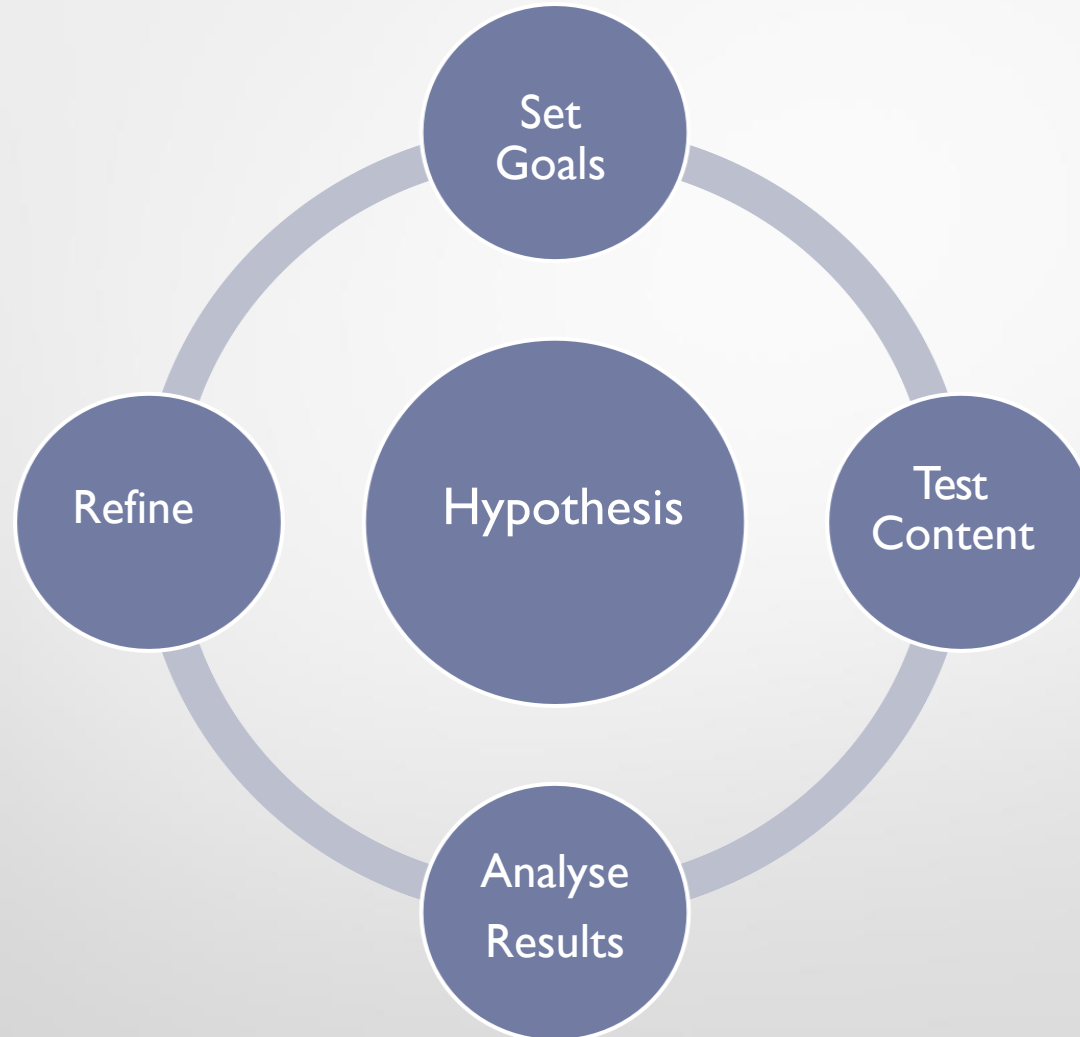
The hypothesis should be considered a living document that is constantly revised and continuously improved as it is tested and new information and/or insights come to light.

It is not permanent and it is not the final answer.

It should be informed by background information, preliminary data analyses, and input from Subject Matter Experts (SME's)

It is often referred to as cause and effect analysis

Overall Process - Iterations



Defining a Hypothesis Statement

Wording a hypothesis statement format

We believe that creating {this feature} for {these people} will achieve {this outcome}. We will validate this when we see {this result} impact our business in {this way} .

For Example

“We believe that creating Zoom lectures for our students will assist them with completing their studies in semester 1 2022. We will validate this when we see that all 98 students can successfully complete ISYS90045 in Semester 1 2022”

Without a(n) hypothesis

What can happen without a hypothesis?

When clients apply bad logic to many important decisions it can lead to undesirable outcomes;

- What we see is an idea that pops into their head, and then somehow it just becomes a fact which occurred when, in a lousy logic moment a CEO stated



“I’ve never heard our customers talk about price, so price doesn’t matter with our products, and I’ve decided we’re going to raise prices.” CEO

Examples: Non Hypothesis vs Hypothesis

Bad Logic



Hypothesis Driven Logic

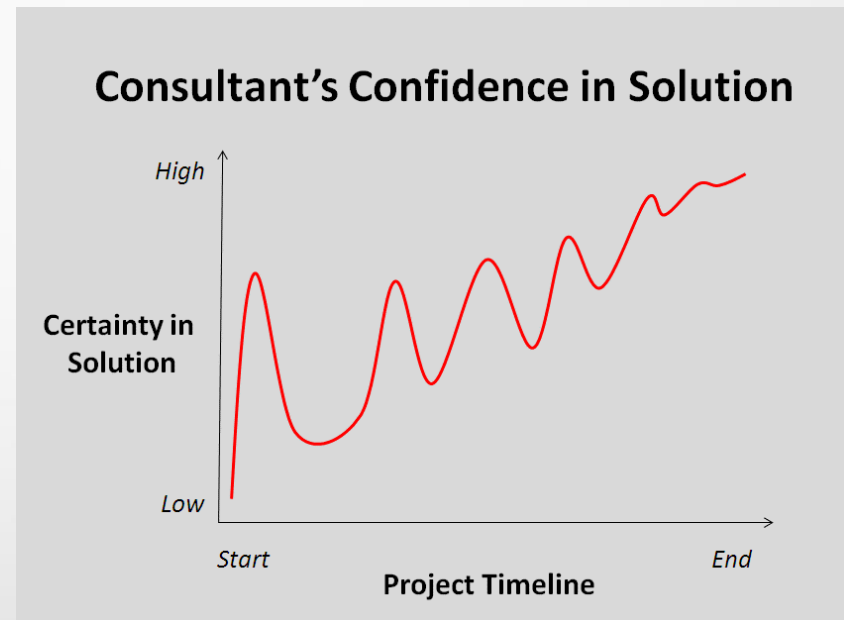


- ▶ Instead of “doing this discount promotion will increase our profits,” a hypothesis-driven approach is to ask “what are the best marketing ideas to increase our profits?” and then conducting a marketing experiment to see which ideas increase profits the most.
- ▶ Instead of “our customers need to have this feature,” ask the question, “what features would our customers value most?” And, then conduct a simple survey having customers rank order the features based on value to them.
- ▶ Instead of “morale is in the toilet because we don’t pay well, so we need to increase pay,” conduct a survey asking, “what is the level of morale?” what are potential issues affecting morale?” and what are the best ideas to improve morale?”

Hypothesis Driven Consulting

Limitations:

- Clients initially may be uncomfortable working this way as it is different from routine tasks and operational process problem solving
- At the start, it may seem as though the Consultant doesn't know what they're doing when they're asking obvious questions and gathering data



Hypothesis Driven Consulting

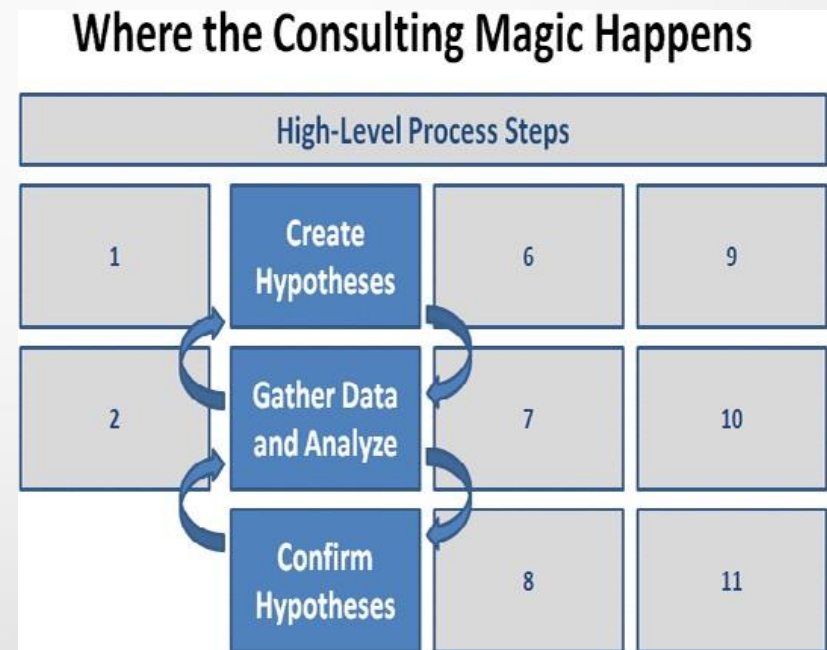
- ▶ Should not be based on **conjecture** or solely on **opinion**
- ▶ Different approach to the linear models of solving problems but it overcomes bottlenecks in waiting for data to be gathered
- ▶ Early, there might not be many concrete facts upon which to base the hypothesis. But the further you get into a client engagement, the more **facts can be gathered to** support or refute your hypothesis
- ▶ NOTE: It is definitely not thinking that you know the answer at the beginning of a client engagement - that attitude creates perceptions that Consultants are **know-it-alls**.
 - ▶ What are some useful tools for gathering information?

Hypothesis Driven Approach

How do we use them?

- ▶ In the generic project plan below, you can see that the hypothesis development happens early on. This typically looks like some best practices research, interviews with clients, and lots of whiteboard time. You put together some initial ideas (logical starting points)

For example



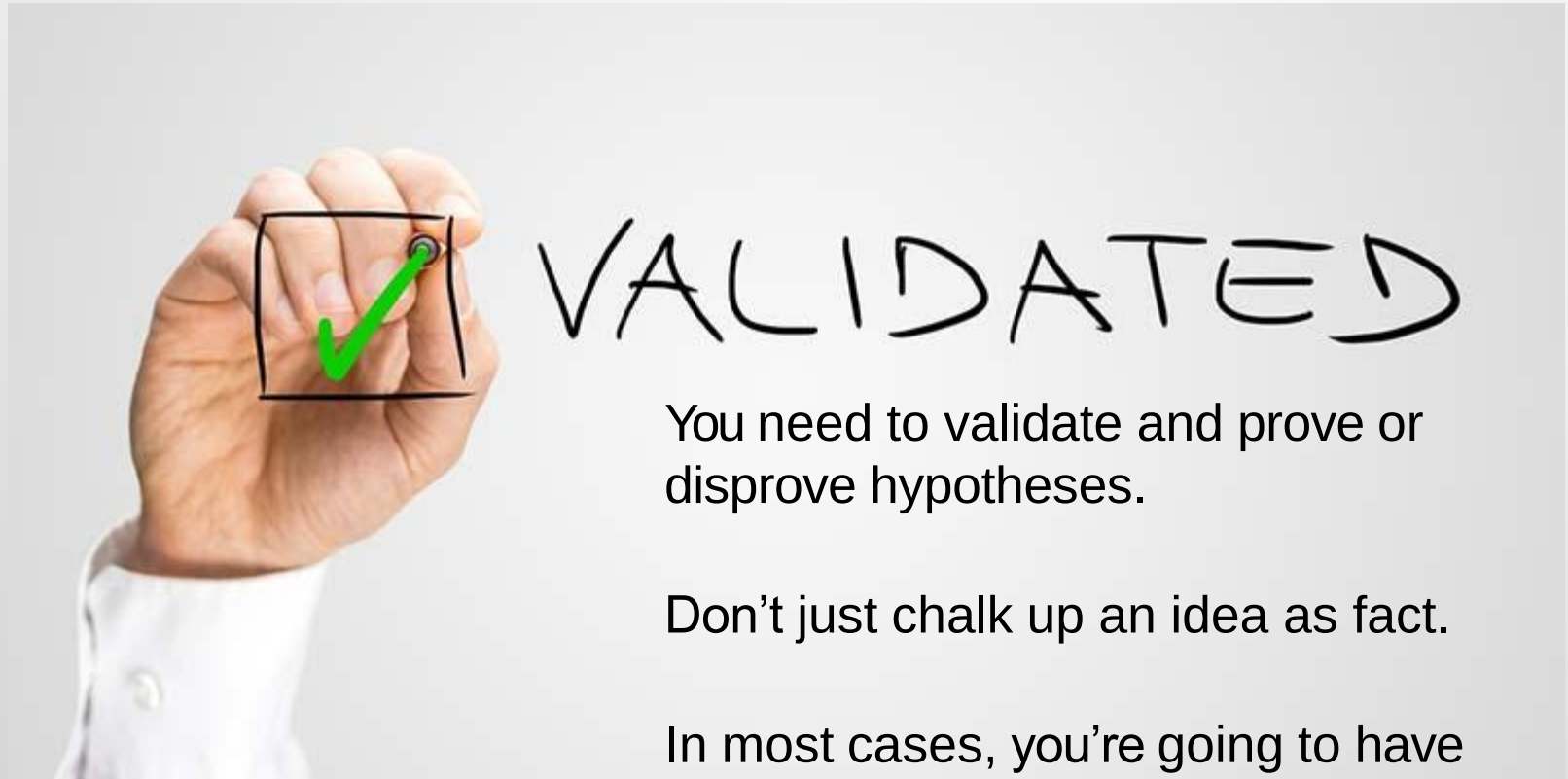
Example Hypothesis Approach

Think broadly about the problem

- ▶ Consultants have the time (clients have day-jobs) and the authority (borrowed influence from their client executives) to cast a wide net for answers.
- ▶ In the example the client might believe ABC is the reason for the fall in market share, but in reality, it is a combination of other reasons that the client did not see (or want to admit). Let's not mistake correlation with causation.



Validate Hypotheses



You need to validate and prove or disprove hypotheses.

Don't just chalk up an idea as fact.

In most cases, you're going to have to create a fact base, through logic, observation, testing, surveys, and analysis

Good Practise for Hypothesis-Driven

- ▶ Be **sensitive** to the fact that most non-consultants are **not** comfortable working this way
 - ▶ Clients may feel you are rushing into a solution without due consideration
- ▶ Inform your hypothesis with all available information including situational **context**
- ▶ **Involve others** - especially your team - in developing and revising your hypothesis
- ▶ **Test** and **improve** your hypothesis constantly as new information becomes available

Not So Good Practice

Don't wait until you have every fact you are seeking - you might be waiting forever and you have timeframes

Don't waste time trying to make your hypothesis and supporting analyses perfect

Don't forget that your hypothesis should be a living document that is continuously improved

Fail to **question** your hypothesis

Defend your hypothesis blindly

Adjust data to fit your hypothesis

Summary

- ▶ Hypothesis based consulting
- ▶ Creating a hypothesis
- ▶ How would we go about testing our hypothesis?
- ▶ The Client and the Hypothesis
- ▶ The importance of facts